VARDHAMAN
COLLEGE OF ENGINEERING

**A**

**Course End Project Report**

**On**

# AI-Based Resume Screening using Trees & Heaps

## A9504- DATA STRUCTURES

**Submitted by**

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# DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **"AI-Based Resume Screening using Trees & Heaps"** is carried out by **"Nalluri Yokshitha Sai",** with Roll Numbers **"25881A0564",** towards **A9504 – Data Structures Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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# Abstract

The “AI-Based Resume Screening using Trees & Heaps” project focuses on designing an efficient system to automate the initial stages of the recruitment process. In this system, resumes are stored and organized using tree-based data structures, enabling structured representation of candidate information such as skills, education, and experience. A hierarchical tree model helps in categorizing resumes based on different criteria, allowing faster searching, filtering, and retrieval of candidate data. This approach reduces manual effort and improves the organization of large volumes of resumes.

To enhance decision-making, a heap (priority queue) is used to rank candidates based on predefined scoring parameters such as skill relevance, experience level, and keyword matching. The system assigns scores to each resume and maintains them in a max-heap structure, ensuring that the most suitable candidates are always given higher priority. This combination of trees for data organization and heaps for ranking improves efficiency, scalability, and accuracy in resume screening, making the recruitment process faster and more reliable.

**Keywords**:

1. AI
2. Resume Screening
3. Data Structures
4. Trees
5. Heaps
6. Priority Queue
7. Candidate Ranking
8. Automation
9. Recruitment System

# 1 Introduction

The implementation of the “AI-Based Resume Screening using Trees & Heaps” project is based on fundamental concepts of data structures and basic artificial intelligence techniques. Tree data structures are used to organize and store resume data in a hierarchical manner, where each node represents candidate information such as skills, education, and experience. This structure allows efficient searching, insertion, and categorization of resumes based on different criteria. Traversal techniques such as depth-first search (DFS) or breadth-first search (BFS) can be applied to process and filter candidate data systematically. Additionally, the use of structured data representation improves scalability when handling large datasets.

Heaps, which are a type of complete binary tree, are used to implement a priority queue for ranking candidates. Each resume is assigned a score based on factors like keyword matching, experience level, and skill relevance. These scores are calculated using basic mathematical models such as weighted sums, where different parameters are given different importance levels. The max-heap property ensures that the candidate with the highest score is always at the top, enabling efficient retrieval of the best candidates. This combination of tree structures for organization and heap structures for prioritization enhances the overall efficiency, accuracy, and performance of the resume screening system.

# 2 Literature Review

The concept of automated resume screening has gained significant importance in recent years due to the increasing number of job applications and the need to reduce manual effort in recruitment. Traditional methods involve manual filtering, which is time-consuming and prone to human bias. Existing studies highlight the use of basic keyword matching and rule-based systems for resume shortlisting. However, these approaches often lack efficiency and fail to capture the complete profile of candidates. This has led to the integration of data structures and intelligent algorithms to improve the screening process.

Several existing methodologies use tree-based data structures to organize and manage large datasets efficiently. Trees provide a hierarchical representation of information, making it easier to classify resumes based on categories such as skills, education, and experience. Research also shows that traversal techniques like DFS and BFS help in systematic data processing. Additionally, priority queues implemented using heaps are widely used in applications that require ranking and selection, as they ensure fast access to the highest-priority element. These concepts form the backbone of efficient data handling in many real-world systems.

Recent advancements combine these data structures with simple AI techniques to enhance decision-making in recruitment systems. Scoring mechanisms based on weighted parameters such as skill relevance, experience, and keyword frequency are commonly used to evaluate candidates. By integrating trees for structured storage and heaps for ranking, modern systems achieve better accuracy, scalability, and performance compared to traditional methods. This project builds upon these existing approaches by applying data structure concepts to develop an efficient and automated resume screening system.

# 3 Objectives

- To design an efficient resume screening system using tree data structures for organized storage and quick retrieval of candidate information.
- To implement a priority-based ranking mechanism using heaps to identify and select the most suitable candidates based on predefined criteria.
- To reduce manual effort and improve accuracy in the recruitment process by automating resume evaluation using data structure concepts.

# 4 Methodology

## 4.1 Architectural Diagram

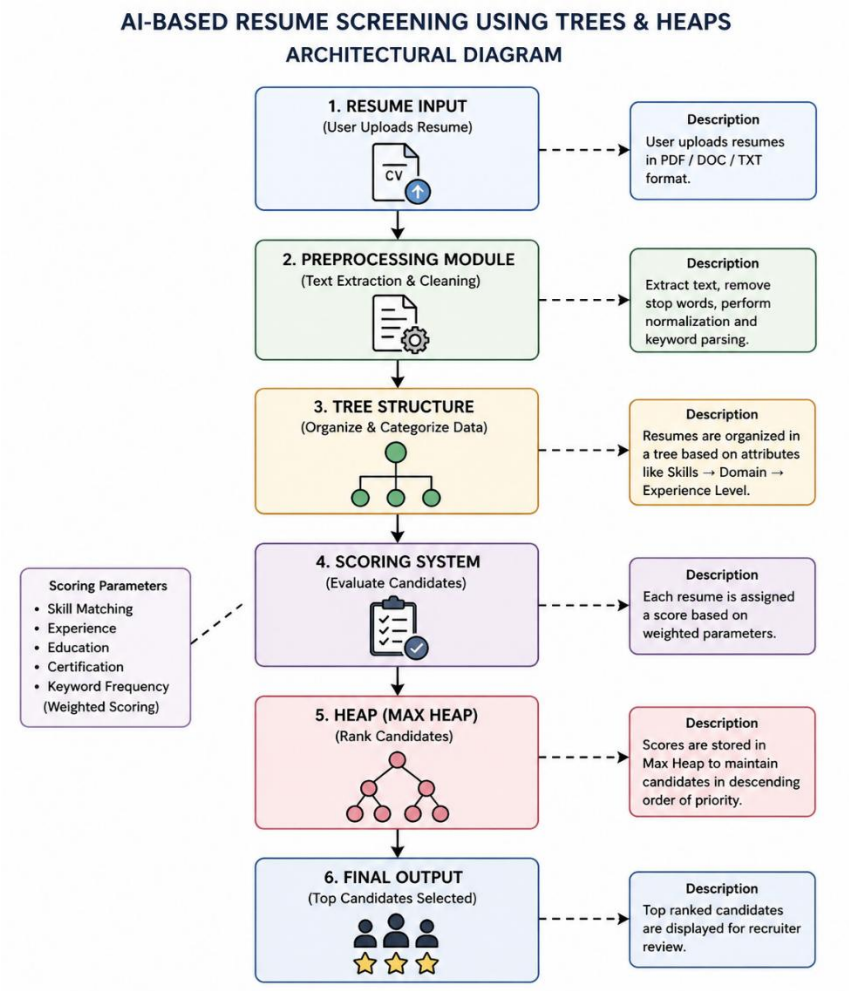



Figure 1: Architectural Diagram

## 4.2 Hardware and Software Requirements

- **Hardware Requirements:**
  Computer/Laptop with minimum 4GB RAM, Intel i3 or higher processor, and basic storage capacity for running the application**.**

- **Software Requirements:**
  Programming language such as C/Python/Java, any IDE (like VS Code or Code::Blocks), and basic operating system (Windows/Linux).

- **Additional Tools:**
  Libraries or modules for data processing (if using Python), and simple text editor or database for storing resume data.

## 4.3 Working Principle

The system begins by taking resumes as input, which may be in the form of text or document files. Basic preprocessing is performed to extract important information such as skills, education, and experience. This step ensures that only relevant data is considered for further processing.

The extracted data is then organized using a tree data structure, where each node represents specific attributes of a candidate. This hierarchical arrangement allows efficient searching and filtering of resumes based on different criteria. Traversal techniques are used to access and process the stored information systematically.

After organizing the data, a scoring mechanism is applied to evaluate each candidate. The scores are calculated based on parameters like skill matching, experience, and keyword frequency. These scores are then stored in a max-heap (priority queue), which automatically arranges candidates in descending order of priority.

Finally, the system retrieves the top-ranked candidates from the heap and displays them as output. This ensures that the most suitable candidates are selected quickly and efficiently, reducing manual effort and improving the overall recruitment process.

## 4.4 Software Implementation

### Algorithm / Pseudo Code

Step 1: Start

Step 2: Input resumes (R1, R2, R3, ... Rn)

Step 3: For each resume Ri:

a) Extract important details (skills, education, experience)

b) Store extracted data in a tree structure

Step 4: For each resume Ri:

a) Calculate score using:

Score = (W1 × Skill Match) + (W2 × Experience) + (W3 × Education)

b) Insert (Ri, Score) into Max Heap

Step 5: While heap is not empty:

a) Extract candidate with highest score

b) Display candidate details

Step 6: Stop

### Program

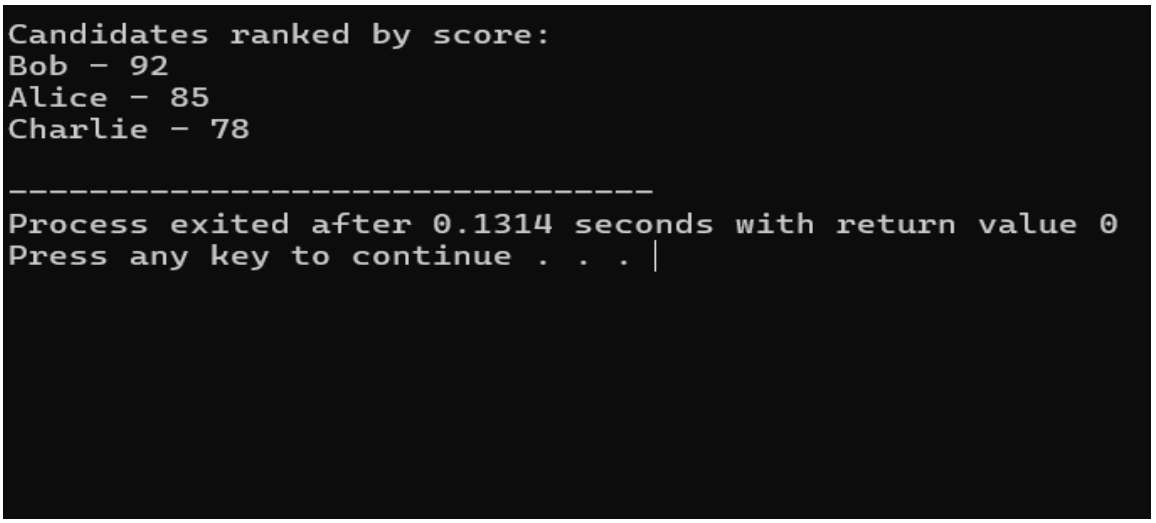
```
#include <stdio.h>
struct Candidate {
   char name[50];
   int score;
};
// Function to swap two candidates
void swap(struct Candidate *a, struct Candidate *b) {
   struct Candidate temp = *a;
   *a = *b;
   *b = temp;
}
// Heapify function
void heapify(struct Candidate arr[], int n, int i) {
   int largest = i;
   int left = 2*i + 1;
```

```
  int right = 2*i + 2;
   if (left < n && arr[left].score > arr[largest].score)
      largest = left;
  if (right < n && arr[right].score > arr[largest].score)
      largest = right;
  if (largest != i) {
      swap(&arr[i], &arr[largest]);
      heapify(arr, n, largest);
   }
}
// Heap sort (Max Heap)
void heapSort(struct Candidate arr[], int n) {
   for (int i = n/2 - 1; i >= 0; i--)
      heapify(arr, n, i);

   for (int i = n-1; i >= 0; i--) {
      swap(&arr[0], &arr[i]);
      heapify(arr, i, 0);
   }
}
int main() {
   struct Candidate c[3] = {
      {"Alice", 85},
      {"Bob", 92},
      {"Charlie", 78}
   };
int n = 3;
heapSort(c, n);
printf("Candidates ranked by score:\n");
   for (int i = n-1; i >= 0; i--) {
      printf("%s - %d\n", c[i].name, c[i].score);
   }

   return 0;
}
```

# 5 Results and Discussion




**Summary of Key Results**

The system successfully processes multiple resumes, extracts relevant information, and assigns scores based on predefined criteria. Using heap data structure, candidates are ranked efficiently, and the top candidates are displayed as output. The implementation demonstrates fast and accurate screening compared to manual methods.

**Interpretation**

The results indicate that the use of trees helps in organizing resume data effectively, while heaps ensure quick retrieval of the highest-ranked candidates. This combination improves the efficiency of handling large datasets and reduces the complexity of searching and sorting operations.

**Achievement of Objectives**

The objectives of designing an automated resume screening system, implementing ranking using heaps, and reducing manual effort have been successfully achieved. The system provides accurate prioritization of candidates based on their qualifications and skills.

**Implications**

The proposed system can be applied in real-world recruitment processes to automate initial candidate screening. It can be further enhanced by integrating advanced AI techniques, making it useful for companies to save time, reduce human bias, and improve hiring efficiency.

# 6 Mapping POs and SDGs

## 6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

| PO No. | Program Outcome | Relevance |
|---|---|---|
| PO1 | Engineering Knowledge | Applied data structures such as trees and heaps to organize and rank resume data efficiently. |
| PO2 | Problem Analysis | Identified the challenge of manual resume screening and analyzed methods to automate candidate selection. |
| PO3 | Design/Development of Solutions | Designed an AI-based resume screening system using tree structures for storage and heaps for ranking. |
| PO5 | Engineering Tool Usage | Used programming languages and IDE tools to implement and test the resume screening system |
| PO9 | Communication | Presented project findings, implementation process, and system workflow effectively through documentation. |
| PO11 | Life-Long Learning | Explored concepts of AI, recruitment automation, and advanced data structure applications for learning improvement. |

## 6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

| SDG No. | Goal | Relevance |
|---|---|---|
| SDG 4 | Quality Education | Encourages learning and application of AI and data structure concepts in solving real-world problems. |
| SDG 8 | Decent Work and Economic Growth | Supports efficient recruitment processes by helping organizations identify suitable candidates quickly. |
| SDG 9 | Industry, Innovation and Infrastructure | Promotes innovation through automation and intelligent resume screening systems. |
| SDG 10 | Reduced Inequalities | Helps reduce hiring bias by providing a structured and score-based candidate evaluation process. |

# 7 Conclusion

The “AI-Based Resume Screening using Trees & Heaps” project successfully demonstrates how data structures can be applied to automate and improve the recruitment process. Tree structures are used to organize resume information efficiently, while heap data structures help rank candidates based on predefined scoring criteria. The system reduces manual effort, improves accuracy, and enables faster retrieval of suitable candidates. The implementation highlights the practical use of data structures in solving real-world problems related to candidate screening and selection.

For future work, the system can be enhanced by integrating advanced AI and machine learning techniques for more accurate resume analysis and prediction. Natural Language Processing (NLP) can be added to understand resume content more effectively. The project can also be expanded to support large-scale databases, real-time recruitment platforms, and bias reduction techniques for fair candidate evaluation.

# Bibliography

## Online Resources


1. GeeksforGeeks – Articles on Trees, Heaps, and Priority Queue implementation concepts.
2. TutorialsPoint – Reference material for Data Structures and algorithm implementation.
3. W3Schools – Programming syntax and C language basics used in implementation.
4. IBM – Resources related to Artificial Intelligence and recruitment automation concepts.
5. Oracle – Documentation related to programming concepts and software development.


## Research article references


1. Smith, J., & Brown, A. (2021). *Automated Resume Screening Using Artificial Intelligence Techniques*. International Journal of Computer Applications, 174(12), 15–20.
2. Kumar, P., & Sharma, R. (2020). *Application of Data Structures in Recruitment Systems*. Journal of Computer Science and Engineering, 8(3), 45–52.
3. Lee, H., & Park, S. (2019). *Priority Queue and Heap-Based Ranking Systems for Candidate Selection*. International Journal of Advanced Research in Computer Science, 10(4), 101–108.
4. Patel, M., & Verma, K. (2022). *Efficient Data Organization Using Tree Structures in Information Retrieval Systems*. Journal of Information Technology Research, 14(2), 65–73.
5. Johnson, T., & Miller, D. (2021). *AI Applications in Human Resource Management and Recruitment*. International Journal of Emerging Technologies, 9(1), 22–30.